

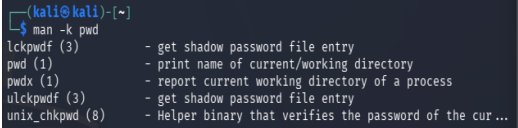
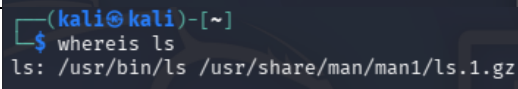
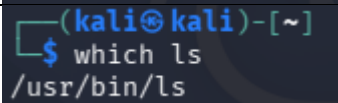
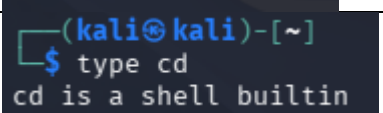
LINUX LAB ASSIGNMENT - II

K V S CHANAKYA | LINUX LAB | 25.08.2026

The main aim of this lab session is to provide hands-on experience on

- Getting Help
- Basic Commands
- Navigation
- File System
- Simple shell script
- Globbing
- Extended Globbing
- Filters -I (head, tail, wc)

1. Getting Help

Task	Command Name	Syntax	Example	Screenshots
To get manual page for the known command	man	man <command>	man ls	
To get manual page for the unknown command	man -k	man -k <keyword>	man -k pwd	
To know the source file binary	whereis	whereis <command>	whereis ls	
To know the path of the command	which	which <command>	which ls	
To know the command is external or internal	type	type <command>	type cd	

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To get help for the internal command	help	help <command>	help cd	<pre>(kali@kali)-[~] └─\$ help cd cd: cd [-L [-P [-e]]] [-@] [dir] Change the shell working directory. Change the current directory to DIR. The default DIR is the value of the HOME shell variable. If DIR is "-", it is converted to \$OLDPWD. The variable CDPATH defines the search path for the directory containing DIR. Alternative directory names in CDPATH are separated by a colon (:). A null directory name is the same as the current directory. If DIR begins with a slash (/), then CDPATH is not used. If the directory is not found, and the shell option `cdable_vars' is set,     the word is assumed to be a variable name.  If that variable has a value     its value is used for DIR.      Options:       -L      force symbolic links to be followed: resolve symbolic               links in DIR after processing instances of `..' -P use the physical directory structure without following symbolic links: resolve symbolic links in DIR before processing instances of `..' -e if the -P option is supplied, and the current working directory cannot be determined successfully, exit with a non-zero status -@ on systems that support it, present a file with extended attributes as a directory containing the file attributes The default is to follow symbolic links, as if `-L' were specified.     `..' is processed by removing the immediately previous pathname component back to a slash or the beginning of DIR. Exit Status: Returns 0 if the directory is changed, and if \$PWD is set successfully wh en -P is used; non-zero otherwise.</pre>
To list out bash commands	compgen	compgen -b	compgen -b	<pre>(kali@kali)-[~] └─\$ compgen -b .: [alias bg bind break builtin caller cd command compgen complete compgen continue declare dirs disown echo</pre>

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To know the usage of the command	-- help	<command> --help	Ls --help	<pre>(kali@kali)~\$ ls --help Usage: ls [OPTION]... [FILE]... List information about the FILEs (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified. Mandatory arguments to long options are mandatory for short options too. -a, --all do not ignore entries starting with . -A, --almost-all do not list implied . and .. --author with -l, print the author of each file -b, --escape print C-style escapes for nongraphic characters --block-size=SIZE with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below -B, --ignore-backups do not list implied entries ending with ~ -c with -lt: sort by, and show, ctime (time of last change of file status information); with -l: show ctime and sort by name; otherwise: sort by ctime, newest first -C list entries by columns --color[=WHEN] color the output WHEN; more info below -d, --directory list directories themselves, not their contents</pre>
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2. Basic Commands

Task	Command Name	Syntax	Example	Screenshots
To know today's date	date	date [option]	date	<pre>(kali@kali)~\$ date Mon Aug 24 12:16:43 PM EDT 2026</pre>
To print calendar	cal	cal	cal	<pre>(kali@kali)~\$ cal August 2026 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31</pre>
To print kernel version	uname	uname -r	uname -r	<pre>(kali@kali)~\$ uname -r 7.0.12+kali-amd64</pre>

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To print default shell	echo	echo \$SHELL	echo \$SHELL	<pre>(kali㉿kali)-[~] \$ echo \$SHELL /usr/bin/zsh</pre>	
To print currently logged in user	whoami	whoami	whoami	<pre>(kali㉿kali)-[~] \$ whoami kali</pre>	
To create shortcut for command	alias	alias name = 'command'	alias list = 'ls -l'	<pre>(kali㉿kali)-[~] \$ alias list='ls -l' (kali㉿kali)-[~] \$ list total 60 drwxr-xr-x 2 kali kali 4096 Aug 11 09:37 Desktop drwxr-xr-x 2 kali kali 4096 Aug 11 09:37 Documents drwxr-xr-x 2 kali kali 4096 Aug 11 09:37 Downloads -rwxrwxr-x 1 kali kali 108 Aug 12 14:39 exitstat.sh drwxrwxr-x 3 kali kali 4096 Aug 22 10:53 Linux drwxr-xr-x 2 kali kali 4096 Aug 11 09:37 Music drwxr-xr-x 2 kali kali 4096 Aug 16 08:01 Pictures drwxr-xr-x 2 kali kali 4096 Aug 11 09:37 Projects drwxr-xr-x 2 kali kali 4096 Aug 11 09:37 Public drwxr-xr-x 2 kali kali 4096 Aug 11 09:37 Templates drwxrwxr-x 2 kali kali 4096 Aug 12 14:34 testdir drwxrwxr-x 2 kali kali 4096 Aug 12 14:41 testdirr drwxrwxr-x 2 kali kali 4096 Aug 13 14:22 test.sh drwxrwxr-x 2 kali kali 4096 Aug 13 14:20 Test.txt drwxr-xr-x 2 kali kali 4096 Aug 11 09:37 Videos</pre>	
To delete shortcut	unalias	unalias name	unalias list	<pre>(kali㉿kali)-[~] \$ unalias list (kali㉿kali)-[~] \$ list Command 'list' not found, did you mean: command 'dist' from deb nmh command 'last' from deb wtmpdb command 'lift' from deb lift command 'hist' from deb loki command 'gist' from deb yorick command 'flist' from deb nmh command 'mlist' from deb mblaze Try: sudo apt install <deb name></pre>	
To change the timestamp of the file	touch	touch [option] file	touch filename	<pre>(kali㉿kali)-[~] \$ touch file1.txt (kali㉿kali)-[~] \$ █</pre>	

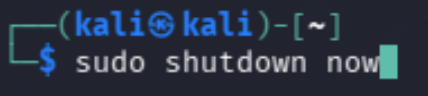
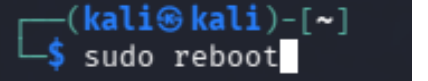
LINUX COMMANDS

To clear the screen	clear	clear	clear	<pre>(kali㉿kali)-[~] \$ clear</pre>	
To create empty files	touch	touch filename.ext	touch file1.txt	<pre>(kali㉿kali)-[~] \$ touch file1.txt (kali㉿kali)-[~] \$</pre>	
To know disk usage	du	du [option] [file]	du -h	<pre>(kali㉿kali)-[~] \$ du -sh du: cannot read directory './Linux/sample_dr': Permission denied 119M .</pre>	
To know free space in the system	df	df [option] [file]	df -h	<pre>(kali㉿kali)-[~] \$ df -h Filesystem Size Used Avail Use% Mounted on udev 869M 0 869M 0% /dev tmpfs 194M 1.2M 193M 1% /run /dev/sda1 79G 17G 58G 23% / tmpfs 967M 4.0K 967M 1% /dev/shm none 1.0M 0 1.0M 0% /run/credentials/systemd-journald tmpfs 967M 80K 967M 1% /tmp none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service tmpfs 194M 108K 194M 1% /run/user/1000</pre>	
To know about the Linux release	lsb_release	lsb_release [option]	lsb_release -a	<pre>(kali㉿kali)-[~] \$ lsb_release -a No LSB modules are available. Distributor ID: Kali Description: Kali GNU/Linux Rolling Release: 2026.3 Codename: kali-rolling</pre>	

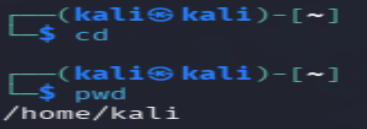
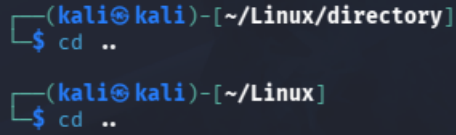
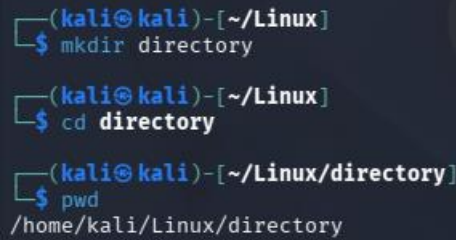
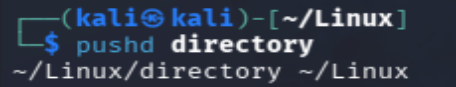
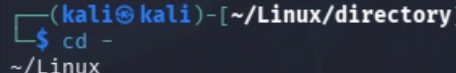
LINUX COMMANDS

To show login history	last	last [option]	last	<pre> (kali@kali)-[~] \$ last kali tty7 :0 Mon Aug 24 12:15 - still logged lightdm tty7 :0 Mon Aug 24 12:10 - 12:15 (00:04) kali tty7 :0 Sat Aug 22 10:50 - still logged lightdm tty7 :0 Sat Aug 22 10:50 - 10:50 (00:00) kali tty7 :0 Wed Aug 19 02:12 - still logged lightdm tty7 :0 Wed Aug 19 02:11 - 02:12 (00:00) kali tty7 :0 Wed Aug 19 02:03 - 02:07 (00:04) lightdm tty7 :0 Wed Aug 19 01:59 - 02:03 (00:03) kali tty7 :0 Sat Aug 15 09:03 - still logged lightdm tty7 :0 Sat Aug 15 09:03 - 09:03 (00:00) kali tty7 :0 Thu Aug 13 14:20 - still logged lightdm tty7 :0 Thu Aug 13 14:20 - 14:20 (00:00) kali tty7 :0 Thu Aug 13 10:06 - still logged lightdm tty7 :0 Thu Aug 13 10:06 - 10:06 (00:00) kali tty7 :0 Wed Aug 12 14:32 - still logged lightdm tty7 :0 Wed Aug 12 14:32 - 14:32 (00:00) postgres pts/2 :0 Tue Aug 11 10:10 - 10:10 (00:00) kali tty7 :0 Tue Aug 11 10:09 - still logged kali tty7 :0 Tue Aug 11 09:37 - still logged lightdm tty7 :0 Tue Aug 11 09:37 - 09:37 (00:00) lightdm tty7 :0 Tue Aug 11 09:29 - still logged postgres pts/2 :0 Tue Jun 16 10:29 - 10:29 (00:00) wtmpd begins Tue Jun 16 10:29:31 2026 </pre>
To check the current user privileges	id	id	id	<pre> (kali@kali)-[~] \$ id uid=1000(kali) gid=1000(kali) groups=1000(kali),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),46(video),46(plugdev),100(users),101(netdev),102(scanner),118(wireshark),119(kaboxer),982(blutetooth),999(l </pre>
To find about the Linux release	cat	cat /etc/os-release	cat /etc/os-release	<pre> (kali@kali)-[~] \$ cat /etc/os-release PRETTY_NAME="Kali GNU/Linux Rolling" NAME="Kali GNU/Linux" VERSION_ID="2026.3" VERSION="2026.3" VERSION_CODENAME=kali-rolling ID=kali ID_LIKE=debian HOME_URL="https://www.kali.org/" SUPPORT_URL="https://forums.kali.org/" BUG_REPORT_URL="https://bugs.kali.org/" ANSI_COLOR="1;31" </pre>
To check system uptime	uptime	uptime	uptime	<pre> (kali@kali)-[~] \$ uptime 12:27:29 up 16 min, 1 user, load average: 0.05, 0.05 </pre>

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To shutdown	shutdown	sudo shutdown [option] [time]	sudo shutdown now	
To restart	reboot	sudo reboot	sudo reboot	

3. Navigation

Task	Command	Syntax	Screenshots
To navigate home directory	cd	cd	
To navigate to the parent directory	cd	cd ..	
To navigate to the child directory	cd	cd foldername	
Alternate command to cd	pushd	pushd directory	
To go back to the previous directory	cd-	cd-	

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To go to the root directory	cd/	cd/	<pre>(kali㉿kali)-[~/Linux] \$ cd / (kali㉿kali)-[/] \$ pwd /</pre>	
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4. File System

Task	Syntax	Command
How to identify the file system	df -T	df -T

```
(kali㉿kali)-[/]
$ df -T
Filesystem      Type      1K-blocks    Used Available Use% Mounted on
udev            devtmpfs   889664         0    889664   0% /dev
tmpfs           tmpfs     197872      1200    196672   1% /run
/dev/sda1       ext4     82083148 17271616  60595984  23% /
tmpfs           tmpfs     989356         4    989352   1% /dev/shm
none            tmpfs      1024         0      1024   0% /run/credentials/systemd-journald.service
tmpfs           tmpfs     989360        40    989320   1% /tmp
none            tmpfs      1024         0      1024   0% /run/credentials/getty@tty1.service
tmpfs           tmpfs     197868       108    197760   1% /run/user/1000
```

Screenshot

5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

Use globbing patterns wherever possible, rather than specifying filenames individually.

```
(kali㉿kali)-[~]
└─$ mkdir practice

(kali㉿kali)-[~]
└─$ cd practice

(kali㉿kali)-[~/practice]
└─$ pwd
/home/kali/practice

(kali㉿kali)-[~/practice]
└─$ ls -ld ~
drwx----- 24 kali kali 4096 Aug 24 13:00 /home/kali

(kali㉿kali)-[~/practice]
└─$ pwd
/home/kali/practice

(kali㉿kali)-[~/practice]
└─$ touch index.html login.html admin.htm report.txt access.log error.log user1.txt user2.log file123 backup1.html backup2.txt

(kali㉿kali)-[~/practice]
└─$ ls
access.log  backup1.html  error.log  index.html  report.txt  user2.log
admin.htm   backup2.txt   file123    login.html  user1.txt
```

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- a. List all files ending in .html.

```
(kali㉿kali)-[~/practice]
$ ls *.html
backup1.html  index.html  login.html
```

- b. List all files other than .html files using extended globbing.

```
(kali㉿kali)-[~/practice]
$ ls !(*.html)
access.log  admin.htm  backup2.txt  error.log  file123  report.txt  user1.txt  user2.log
```

- c. List filenames containing at least one numeric digit.

```
(kali㉿kali)-[~/practice]
$ ls *[0-9]*
backup1.html  backup2.txt  file123  user1.txt  user2.log
```

- d. List files whose names begin with user followed by a single digit.

```
(kali㉿kali)-[~/practice]
$ ls user?.*
user1.txt  user2.log
```

- e. List files ending in either .txt or .log using a single extended glob pattern.

```
(kali㉿kali)-[~/practice]
$ ls *.(txt|log)
access.log  backup2.txt  error.log  report.txt  user1.txt  user2.log
```

- f. List files that do not end in .txt or .log.

```
(kali@kali)-[~/practice]
$ ls !(*.txt|*.log)
admin.htm  backup1.html  file123  index.html  login.html
```

- g. List files beginning with either access or error using extended globbing.

```
(kali@kali)-[~/practice]
$ ls @(access|error).log
access.log  error.log
```

- h. List .html files whose filenames contain a numeric digit.

```
(kali@kali)-[~/practice]
$ ls *[0-9]*.html
backup1.html
```

6. Extended Globbing

- a. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

Write an extended glob pattern that selects only access.log, error.log, and auth.log.

Select all files except notes.txt.

Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.

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```
(kali@kali)-[~/practice]
$ mkdir practice1

(kali@kali)-[~/practice]
$ cd practice1

(kali@kali)-[~/practice/practice1]
$ touch access.log error.log auth.log auth.log.1 auth.log.2 system.log notes.txt

(kali@kali)-[~/practice/practice1]
$ shopt -s extglob

(kali@kali)-[~/practice/practice1]
$ ls auth.log.@(1|2)
auth.log.1  auth.log.2

(kali@kali)-[~/practice/practice1]
$ ls @(access|error|auth).log
access.log  auth.log  error.log

(kali@kali)-[~/practice/practice1]
$ ls !(notes.txt)
access.log  auth.log  auth.log.1  auth.log.2  error.log  system.log

(kali@kali)-[~/practice/practice1]
$ ls auth.log.@(1|2)
auth.log.1  auth.log.2
```

b. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log

construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.

```
(kali@kali)-[~/practice/practice1]
$ touch malware1.log malware2.log malwareA.log malware_report.txt, normal.log

(kali@kali)-[~/practice/practice1]
$ shopt -s extglob

(kali@kali)-[~/practice/practice1]
$ ls malware*[0-9]*.log
malware1.log  malware2.log
```

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c. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log

use extended globbing to select .log files containing either failed_login or successful_login Construct a pattern that lists all .log files except firewall.log.

```
(kali㉿kali)-[~/practice/practice1]
$ touch failed_login.log successful_login.log login_backup.log login.txt firewall.log

(kali㉿kali)-[~/practice/practice1]
$ shopt -s extglob

(kali㉿kali)-[~/practice/practice1]
$ ls @(failed_login|successful_login).log
failed_login.log  successful_login.log

(kali㉿kali)-[~/practice/practice1]
$ ls !("firewall.log")
access.log  auth.log.2      login_backup.log  malware2.log  notes.txt
auth.log    error.log       login.txt         malwareA.log  successful_login.log
auth.log.1  failed_login.log  malware1.log     normal.log    system.log
```

7. Filters

Create files named as login.log, access.log, security.log, and incident.log in a directory and answer the questions below:

a. Display the first 10 entries of login.log.

```
(kali㉿kali)-[~/filters]
$ head -10 login.log
Aug 15 09:00:00 server sshd[2101]: Accepted password for student1 from 192.168.10.21
Aug 15 09:01:07 server sshd[2102]: Failed password for student2 from 198.51.100.44
Aug 15 09:02:14 server sshd[2103]: Failed password for analyst from 203.0.113.77
Aug 15 09:03:21 server sshd[2104]: Accepted password for admin from 192.168.10.45
Aug 15 09:04:28 server sshd[2105]: Accepted password for root from 192.168.10.21
Aug 15 09:05:35 server sshd[2106]: Failed password for guest from 198.51.100.44
Aug 15 09:06:42 server sshd[2107]: Failed password for student1 from 203.0.113.77
Aug 15 09:07:49 server sshd[2108]: Accepted password for student2 from 192.168.10.45
Aug 15 09:08:56 server sshd[2109]: Accepted password for analyst from 192.168.10.21
Aug 15 09:09:03 server sshd[2110]: Failed password for admin from 198.51.100.44
```

b. Display the first 5 entries of access.log.

```
(kali㉿kali)-[~/filters]
$ head -5 access.log
192.168.10.21 - - [15/Aug/2026:09:00:00 +0530] "GET / HTTP/1.1" 200 512
192.168.10.22 - - [15/Aug/2026:09:00:37 +0530] "POST /login HTTP/1.1" 200 649
192.168.10.30 - - [15/Aug/2026:09:01:14 +0530] "GET /dashboard HTTP/1.1" 302 786
192.168.10.45 - - [15/Aug/2026:09:01:51 +0530] "GET /admin HTTP/1.1" 403 923
203.0.113.25 - - [15/Aug/2026:09:02:28 +0530] "GET /robots.txt HTTP/1.1" 404 1060
```

c. Display the first 15 security events from security.log.

```
(kali㉿kali)-[~/filters]
$ head -15 security.log
2026-08-15 09:00:00 INFO user login successful; source=192.168.10.21; event_id=SEC0001
2026-08-15 09:00:29 WARN failed login attempt; source=192.168.10.22; event_id=SEC0002
2026-08-15 09:00:58 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0003
2026-08-15 09:01:27 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0004
2026-08-15 09:01:56 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0005
2026-08-15 09:02:25 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0006
2026-08-15 09:02:54 INFO user logout; source=203.0.113.77; event_id=SEC0007
2026-08-15 09:03:23 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0008
2026-08-15 09:03:52 INFO user login successful; source=192.168.10.21; event_id=SEC0009
2026-08-15 09:04:21 WARN failed login attempt; source=192.168.10.22; event_id=SEC0010
2026-08-15 09:04:50 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0011
2026-08-15 09:05:19 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0012
2026-08-15 09:05:48 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0013
2026-08-15 09:06:17 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0014
2026-08-15 09:06:46 INFO user logout; source=203.0.113.77; event_id=SEC0015
```

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- d. Display the first 20 events recorded in incident.log.

```
(kali㉿kali)-[~/filters]
$ head -20 incident.log
2026-08-15 09:00:00 INFO login success; source=192.168.10.21; incident_event=001
2026-08-15 09:00:13 INFO web request; source=192.168.10.22; incident_event=002
2026-08-15 09:00:26 INFO logout; source=192.168.10.30; incident_event=003
2026-08-15 09:00:39 INFO scheduled scan; source=192.168.10.45; incident_event=004
2026-08-15 09:00:52 INFO access check; source=203.0.113.25; incident_event=005
2026-08-15 09:01:05 INFO login success; source=198.51.100.44; incident_event=006
2026-08-15 09:01:18 INFO web request; source=203.0.113.77; incident_event=007
2026-08-15 09:01:31 INFO logout; source=198.51.100.91; incident_event=008
2026-08-15 09:01:44 INFO scheduled scan; source=192.168.10.21; incident_event=009
2026-08-15 09:01:57 INFO access check; source=192.168.10.22; incident_event=010
2026-08-15 09:02:10 INFO login success; source=192.168.10.30; incident_event=011
2026-08-15 09:02:23 INFO web request; source=192.168.10.45; incident_event=012
2026-08-15 09:02:36 INFO logout; source=203.0.113.25; incident_event=013
2026-08-15 09:02:49 INFO scheduled scan; source=198.51.100.44; incident_event=014
2026-08-15 09:03:02 INFO access check; source=203.0.113.77; incident_event=015
2026-08-15 09:03:15 INFO login success; source=198.51.100.91; incident_event=016
2026-08-15 09:03:28 INFO web request; source=192.168.10.21; incident_event=017
2026-08-15 09:03:41 INFO logout; source=192.168.10.22; incident_event=018
2026-08-15 09:03:54 INFO scheduled scan; source=192.168.10.30; incident_event=019
2026-08-15 09:04:07 INFO access check; source=192.168.10.45; incident_event=020
```

- e. Display only the first entry in login.log.

```
(kali㉿kali)-[~/filters]
$ head -1 login.log
Aug 15 09:00:00 server sshd[2101]: Accepted password for student1 from 192.168.10.21
```


LINUX COMMANDS

- f. Display the first 25 web requests recorded in access.log.

```
(kali@kali)-[~/filters]
$ head -25 access.log
192.168.10.21 - - [15/Aug/2026:09:00:00 +0530] "GET / HTTP/1.1" 200 512
192.168.10.22 - - [15/Aug/2026:09:00:37 +0530] "POST /login HTTP/1.1" 200 649
192.168.10.30 - - [15/Aug/2026:09:01:14 +0530] "GET /dashboard HTTP/1.1" 302 786
192.168.10.45 - - [15/Aug/2026:09:01:51 +0530] "GET /admin HTTP/1.1" 403 923
203.0.113.25 - - [15/Aug/2026:09:02:28 +0530] "GET /robots.txt HTTP/1.1" 404 1060
198.51.100.44 - - [15/Aug/2026:09:03:05 +0530] "POST /api/status HTTP/1.1" 401 1197
203.0.113.77 - - [15/Aug/2026:09:03:42 +0530] "GET /assets/app.js HTTP/1.1" 200 1334
198.51.100.91 - - [15/Aug/2026:09:04:19 +0530] "GET / HTTP/1.1" 200 1471
192.168.10.21 - - [15/Aug/2026:09:04:56 +0530] "GET /login HTTP/1.1" 302 1608
192.168.10.22 - - [15/Aug/2026:09:05:33 +0530] "POST /dashboard HTTP/1.1" 403 1745
192.168.10.30 - - [15/Aug/2026:09:06:10 +0530] "GET /admin HTTP/1.1" 404 1882
192.168.10.45 - - [15/Aug/2026:09:06:47 +0530] "GET /robots.txt HTTP/1.1" 401 2019
203.0.113.25 - - [15/Aug/2026:09:07:24 +0530] "GET /api/status HTTP/1.1" 200 2156
198.51.100.44 - - [15/Aug/2026:09:08:01 +0530] "POST /assets/app.js HTTP/1.1" 200 2293
203.0.113.77 - - [15/Aug/2026:09:08:38 +0530] "GET / HTTP/1.1" 302 2430
198.51.100.91 - - [15/Aug/2026:09:09:15 +0530] "GET /login HTTP/1.1" 403 2567
192.168.10.21 - - [15/Aug/2026:09:09:52 +0530] "GET /dashboard HTTP/1.1" 404 2704
192.168.10.22 - - [15/Aug/2026:09:10:29 +0530] "POST /admin HTTP/1.1" 401 2841
192.168.10.30 - - [15/Aug/2026:09:11:06 +0530] "GET /robots.txt HTTP/1.1" 200 2978
192.168.10.45 - - [15/Aug/2026:09:11:43 +0530] "GET /api/status HTTP/1.1" 200 3115
203.0.113.25 - - [15/Aug/2026:09:12:20 +0530] "GET /assets/app.js HTTP/1.1" 302 3252
198.51.100.44 - - [15/Aug/2026:09:12:57 +0530] "POST / HTTP/1.1" 403 3389
203.0.113.77 - - [15/Aug/2026:09:13:34 +0530] "GET /login HTTP/1.1" 404 3526
198.51.100.91 - - [15/Aug/2026:09:14:11 +0530] "GET /dashboard HTTP/1.1" 401 3663
192.168.10.21 - - [15/Aug/2026:09:14:48 +0530] "GET /admin HTTP/1.1" 403 3800
```

- g. Display the last 10 entries of login.log.

```
(kali@kali)-[~/filters]
$ tail -10 login.log
Aug 15 09:40:40 server sshd[2141]: Accepted password for root from 192.168.10.21
Aug 15 09:41:47 server sshd[2142]: Failed password for guest from 198.51.100.44
Aug 15 09:42:54 server sshd[2143]: Failed password for student1 from 203.0.113.77
Aug 15 09:43:01 server sshd[2144]: Accepted password for student2 from 192.168.10.45
Aug 15 09:44:08 server sshd[2145]: Accepted password for analyst from 192.168.10.21
Aug 15 09:45:15 server sshd[2146]: Failed password for admin from 198.51.100.44
Aug 15 09:46:22 server sshd[2147]: Failed password for root from 203.0.113.77
Aug 15 09:47:29 server sshd[2148]: Accepted password for guest from 192.168.10.45
Aug 15 09:48:36 server sshd[2149]: Accepted password for student1 from 192.168.10.21
Aug 15 09:49:43 server sshd[2150]: Failed password for student2 from 198.51.100.44
```

LINUX COMMANDS

- h. Display the last 5 entries of access.log.

```
(kali@kali)-[~/filters]
$ tail -5 access.log
192.168.10.45 - - [15/Aug/2026:09:46:15 +0530] "GET /api/status HTTP/1.1" 403 2595
203.0.113.25 - - [15/Aug/2026:09:46:52 +0530] "GET /assets/app.js HTTP/1.1" 404 2732
198.51.100.44 - - [15/Aug/2026:09:47:29 +0530] "POST / HTTP/1.1" 401 2869
203.0.113.77 - - [15/Aug/2026:09:48:06 +0530] "GET /login HTTP/1.1" 200 3006
198.51.100.91 - - [15/Aug/2026:09:48:43 +0530] "GET /dashboard HTTP/1.1" 200 3143
```

- i. Display the 20 most recent security events from security.log.

```
(kali@kali)-[~/filters]
$ tail -20 security.log
2026-08-15 09:48:20 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0101
2026-08-15 09:48:49 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0102
2026-08-15 09:49:18 INFO user logout; source=203.0.113.77; event_id=SEC0103
2026-08-15 09:49:47 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0104
2026-08-15 09:50:16 INFO user login successful; source=192.168.10.21; event_id=SEC0105
2026-08-15 09:50:45 WARN failed login attempt; source=192.168.10.22; event_id=SEC0106
2026-08-15 09:51:14 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0107
2026-08-15 09:51:43 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0108
2026-08-15 09:52:12 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0109
2026-08-15 09:52:41 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0110
2026-08-15 09:53:10 INFO user logout; source=203.0.113.77; event_id=SEC0111
2026-08-15 09:53:39 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0112
2026-08-15 09:54:08 INFO user login successful; source=192.168.10.21; event_id=SEC0113
2026-08-15 09:54:37 WARN failed login attempt; source=192.168.10.22; event_id=SEC0114
2026-08-15 09:55:06 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0115
2026-08-15 09:55:35 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0116
2026-08-15 09:56:04 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0117
2026-08-15 09:56:33 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0118
2026-08-15 09:57:02 INFO user logout; source=203.0.113.77; event_id=SEC0119
2026-08-15 09:57:31 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0120
```

- j. Display the 25 most recent events from incident.log.

```
(kali㉿kali)-[~/filters]
$ tail -25 incident.log
2026-08-15 10:42:55 INFO login success; source=192.168.10.45; incident_event=476
2026-08-15 10:43:08 INFO web request; source=203.0.113.25; incident_event=477
2026-08-15 10:43:21 INFO logout; source=198.51.100.44; incident_event=478
2026-08-15 10:43:34 INFO scheduled scan; source=203.0.113.77; incident_event=479
2026-08-15 10:43:47 INFO access check; source=198.51.100.91; incident_event=480
2026-08-15 10:44:00 INFO login success; source=192.168.10.21; incident_event=481
2026-08-15 10:44:13 INFO web request; source=192.168.10.22; incident_event=482
2026-08-15 10:44:26 INFO logout; source=192.168.10.30; incident_event=483
2026-08-15 10:44:39 INFO scheduled scan; source=192.168.10.45; incident_event=484
2026-08-15 10:44:52 INFO access check; source=203.0.113.25; incident_event=485
2026-08-15 10:45:05 INFO login success; source=198.51.100.44; incident_event=486
2026-08-15 10:45:18 INFO web request; source=203.0.113.77; incident_event=487
2026-08-15 10:45:31 INFO logout; source=198.51.100.91; incident_event=488
2026-08-15 10:45:44 INFO scheduled scan; source=192.168.10.21; incident_event=489
2026-08-15 10:45:57 INFO access check; source=192.168.10.22; incident_event=490
2026-08-15 10:46:10 INFO login success; source=192.168.10.30; incident_event=491
2026-08-15 10:46:23 INFO web request; source=192.168.10.45; incident_event=492
2026-08-15 10:46:36 INFO logout; source=203.0.113.25; incident_event=493
2026-08-15 10:46:49 INFO scheduled scan; source=198.51.100.44; incident_event=494
2026-08-15 10:47:02 INFO access check; source=203.0.113.77; incident_event=495
2026-08-15 10:47:15 INFO login success; source=198.51.100.91; incident_event=496
2026-08-15 10:47:28 INFO web request; source=192.168.10.21; incident_event=497
2026-08-15 10:47:41 INFO logout; source=192.168.10.22; incident_event=498
2026-08-15 10:47:54 INFO scheduled scan; source=192.168.10.30; incident_event=499
2026-08-15 10:48:07 INFO access check; source=192.168.10.45; incident_event=500
```

LINUX COMMANDS

- k. Display only the last entry in security.log.

```
(kali㉿kali)-[~/filters]
$ tail -1 security.log
2026-08-15 09:57:31 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0120
```

- l. Display the last 50 events recorded in incident.log.

```
(kali㉿kali)-[~/filters]
$ tail -50 incident.log
2026-08-15 10:37:30 INFO login success; source=192.168.10.30; incident_event=451
2026-08-15 10:37:43 INFO web request; source=192.168.10.45; incident_event=452
2026-08-15 10:37:56 INFO logout; source=203.0.113.25; incident_event=453
2026-08-15 10:38:09 INFO scheduled scan; source=198.51.100.44; incident_event=454
2026-08-15 10:38:22 INFO access check; source=203.0.113.77; incident_event=455
2026-08-15 10:38:35 INFO login success; source=198.51.100.91; incident_event=456
2026-08-15 10:38:48 INFO web request; source=192.168.10.21; incident_event=457
2026-08-15 10:39:01 INFO logout; source=192.168.10.22; incident_event=458
2026-08-15 10:39:14 INFO scheduled scan; source=192.168.10.30; incident_event=459
2026-08-15 10:39:27 INFO access check; source=192.168.10.45; incident_event=460
2026-08-15 10:39:40 INFO login success; source=203.0.113.25; incident_event=461
2026-08-15 10:39:53 INFO web request; source=198.51.100.44; incident_event=462
2026-08-15 10:40:06 INFO logout; source=203.0.113.77; incident_event=463
2026-08-15 10:40:19 INFO scheduled scan; source=198.51.100.91; incident_event=464
2026-08-15 10:40:32 INFO access check; source=192.168.10.21; incident_event=465
2026-08-15 10:40:45 INFO login success; source=192.168.10.22; incident_event=466
2026-08-15 10:40:58 INFO web request; source=192.168.10.30; incident_event=467
2026-08-15 10:41:11 INFO logout; source=192.168.10.45; incident_event=468
2026-08-15 10:41:24 INFO scheduled scan; source=203.0.113.25; incident_event=469
2026-08-15 10:41:37 INFO access check; source=198.51.100.44; incident_event=470
2026-08-15 10:41:50 INFO login success; source=203.0.113.77; incident_event=471
2026-08-15 10:42:03 INFO web request; source=198.51.100.91; incident_event=472
2026-08-15 10:42:16 INFO logout; source=192.168.10.21; incident_event=473
2026-08-15 10:42:29 INFO scheduled scan; source=192.168.10.22; incident_event=474
2026-08-15 10:42:42 INFO access check; source=192.168.10.30; incident_event=475
2026-08-15 10:42:55 INFO login success; source=192.168.10.45; incident_event=476
2026-08-15 10:43:08 INFO web request; source=203.0.113.25; incident_event=477
2026-08-15 10:43:21 INFO logout; source=198.51.100.44; incident_event=478
2026-08-15 10:43:34 INFO scheduled scan; source=203.0.113.77; incident_event=479
2026-08-15 10:43:47 INFO access check; source=198.51.100.91; incident_event=480
2026-08-15 10:44:00 INFO login success; source=192.168.10.21; incident_event=481
2026-08-15 10:44:13 INFO web request; source=192.168.10.22; incident_event=482
2026-08-15 10:44:26 INFO logout; source=192.168.10.30; incident_event=483
2026-08-15 10:44:39 INFO scheduled scan; source=192.168.10.45; incident_event=484
2026-08-15 10:44:52 INFO access check; source=203.0.113.25; incident_event=485
2026-08-15 10:45:05 INFO login success; source=198.51.100.44; incident_event=486
2026-08-15 10:45:18 INFO web request; source=203.0.113.77; incident_event=487
2026-08-15 10:45:31 INFO logout; source=198.51.100.91; incident_event=488
2026-08-15 10:45:44 INFO scheduled scan; source=192.168.10.21; incident_event=489
2026-08-15 10:45:57 INFO access check; source=192.168.10.22; incident_event=490
2026-08-15 10:46:10 INFO login success; source=192.168.10.30; incident_event=491
2026-08-15 10:46:23 INFO web request; source=192.168.10.45; incident_event=492
2026-08-15 10:46:36 INFO logout; source=203.0.113.25; incident_event=493
2026-08-15 10:46:49 INFO scheduled scan; source=198.51.100.44; incident_event=494
2026-08-15 10:47:02 INFO access check; source=203.0.113.77; incident_event=495
```


- m. Display the contents of login.log starting from line 41 until the end of the file.

```
(kali@kali)-[~/filters]
$ tail -n +41 login.log
Aug 15 09:40:40 server sshd[2141]: Accepted password for root from 192.168.10.21
Aug 15 09:41:47 server sshd[2142]: Failed password for guest from 198.51.100.44
Aug 15 09:42:54 server sshd[2143]: Failed password for student1 from 203.0.113.77
Aug 15 09:43:01 server sshd[2144]: Accepted password for student2 from 192.168.10.45
Aug 15 09:44:08 server sshd[2145]: Accepted password for analyst from 192.168.10.21
Aug 15 09:45:15 server sshd[2146]: Failed password for admin from 198.51.100.44
Aug 15 09:46:22 server sshd[2147]: Failed password for root from 203.0.113.77
Aug 15 09:47:29 server sshd[2148]: Accepted password for guest from 192.168.10.45
Aug 15 09:48:36 server sshd[2149]: Accepted password for student1 from 192.168.10.21
Aug 15 09:49:43 server sshd[2150]: Failed password for student2 from 198.51.100.44
```

- n. Display the contents of security.log starting from line 101 until the end of the file.

```
(kali@kali)-[~/filters]
$ tail -n +101 security.log
2026-08-15 09:48:20 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0101
2026-08-15 09:48:49 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0102
2026-08-15 09:49:18 INFO user logout; source=203.0.113.77; event_id=SEC0103
2026-08-15 09:49:47 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0104
2026-08-15 09:50:16 INFO user login successful; source=192.168.10.21; event_id=SEC0105
2026-08-15 09:50:45 WARN failed login attempt; source=192.168.10.22; event_id=SEC0106
2026-08-15 09:51:14 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0107
2026-08-15 09:51:43 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0108
2026-08-15 09:52:12 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0109
2026-08-15 09:52:41 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0110
2026-08-15 09:53:10 INFO user logout; source=203.0.113.77; event_id=SEC0111
2026-08-15 09:53:39 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0112
2026-08-15 09:54:08 INFO user login successful; source=192.168.10.21; event_id=SEC0113
2026-08-15 09:54:37 WARN failed login attempt; source=192.168.10.22; event_id=SEC0114
2026-08-15 09:55:06 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0115
2026-08-15 09:55:35 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0116
2026-08-15 09:56:04 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0117
2026-08-15 09:56:33 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0118
2026-08-15 09:57:02 INFO user logout; source=203.0.113.77; event_id=SEC0119
2026-08-15 09:57:31 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0120
```

- o. Find the total number of lines in login.log.

```
(kali㉿kali)-[~/filters]
$ wc -l login.log
50 login.log
```

- p. Find the total number of lines in access.log.

```
(kali㉿kali)-[~/filters]
$ wc -l access.log
80 access.log
```

- q. Find the total number of security events in security.log.

```
(kali㉿kali)-[~/filters]
$ wc -l security.log
120 security.log
```

- r. Determine the total number of events recorded in incident.log.

```
(kali㉿kali)-[~/filters]
$ wc -l incident.log
500 incident.log
```

- s. Count the number of words in login.log.

```
(kali㉿kali)-[~/filters]
$ wc -w login.log
550 login.log
```

- t. Count the number of bytes in security.log.

```
(kali㉿kali)-[~/filters]
$ wc -c security.log
10875 security.log
```

LINUX COMMANDS

- u. Display the line, word, and byte counts of access.log.

```
(kali㉿kali)-[~/filters]
$ wc access.log
 80  800 6414 access.log
```

- v. Find the line counts of all four .log files using a single wc command.

```
(kali㉿kali)-[~/filters]
$ wc -l login.log access.log security.log incident.log
 50 login.log
 80 access.log
120 security.log
500 incident.log
750 total
```

- w. Display the first 20 entries of login.log and, from those entries, display only the last 5.

```
(kali㉿kali)-[~/filters]
$ head -20 login.log | tail -5
Aug 15 09:15:45 server sshd[2116]: Accepted password for admin from 192.168.10.45
Aug 15 09:16:52 server sshd[2117]: Accepted password for root from 192.168.10.21
Aug 15 09:17:59 server sshd[2118]: Failed password for guest from 198.51.100.44
Aug 15 09:18:06 server sshd[2119]: Failed password for student1 from 203.0.113.77
Aug 15 09:19:13 server sshd[2120]: Accepted password for student2 from 192.168.10.45
```

- x. Display the last 20 entries of login.log and, from those entries, display only the first 5.

```
(kali㉿kali)-[~/filters]
$ tail -20 login.log | head -5
Aug 15 09:30:30 server sshd[2131]: Failed password for student1 from 203.0.113.77
Aug 15 09:31:37 server sshd[2132]: Accepted password for student2 from 192.168.10.45
Aug 15 09:32:44 server sshd[2133]: Accepted password for analyst from 192.168.10.21
Aug 15 09:33:51 server sshd[2134]: Failed password for admin from 198.51.100.44
Aug 15 09:34:58 server sshd[2135]: Failed password for root from 203.0.113.77
```

LINUX COMMANDS

- y. Display the first 30 entries of access.log and then display only the last 10 entries from that output.

```
(kali@kali)-[~/filters]
$ head -30 access.log | tail -10
203.0.113.25 - - [15/Aug/2026:09:12:20 +0530] "GET /assets/app.js HTTP/1.1" 302 3252
198.51.100.44 - - [15/Aug/2026:09:12:57 +0530] "POST / HTTP/1.1" 403 3389
203.0.113.77 - - [15/Aug/2026:09:13:34 +0530] "GET /login HTTP/1.1" 404 3526
198.51.100.91 - - [15/Aug/2026:09:14:11 +0530] "GET /dashboard HTTP/1.1" 401 3663
192.168.10.21 - - [15/Aug/2026:09:14:48 +0530] "GET /admin HTTP/1.1" 200 3800
192.168.10.22 - - [15/Aug/2026:09:15:25 +0530] "POST /robots.txt HTTP/1.1" 200 3937
192.168.10.30 - - [15/Aug/2026:09:16:02 +0530] "GET /api/status HTTP/1.1" 302 4074
192.168.10.45 - - [15/Aug/2026:09:16:39 +0530] "GET /assets/app.js HTTP/1.1" 403 4211
203.0.113.25 - - [15/Aug/2026:09:17:16 +0530] "GET / HTTP/1.1" 404 4348
198.51.100.44 - - [15/Aug/2026:09:17:53 +0530] "POST /login HTTP/1.1" 401 4485
```

- z. Display the last 30 entries of access.log and then display only the first 10 entries.

```
(kali@kali)-[~/filters]
$ tail -30 access.log | head -10
192.168.10.30 - - [15/Aug/2026:09:30:50 +0530] "GET /login HTTP/1.1" 302 3266
192.168.10.45 - - [15/Aug/2026:09:31:27 +0530] "GET /dashboard HTTP/1.1" 403 3403
203.0.113.25 - - [15/Aug/2026:09:32:04 +0530] "GET /admin HTTP/1.1" 404 3540
198.51.100.44 - - [15/Aug/2026:09:32:41 +0530] "POST /robots.txt HTTP/1.1" 401 3677
203.0.113.77 - - [15/Aug/2026:09:33:18 +0530] "GET /api/status HTTP/1.1" 200 3814
198.51.100.91 - - [15/Aug/2026:09:33:55 +0530] "GET /assets/app.js HTTP/1.1" 200 3951
192.168.10.21 - - [15/Aug/2026:09:34:32 +0530] "GET / HTTP/1.1" 302 4088
192.168.10.22 - - [15/Aug/2026:09:35:09 +0530] "POST /login HTTP/1.1" 403 4225
192.168.10.30 - - [15/Aug/2026:09:35:46 +0530] "GET /dashboard HTTP/1.1" 404 4362
192.168.10.45 - - [15/Aug/2026:09:36:23 +0530] "GET /admin HTTP/1.1" 401 4499
```

- aa. Display the first 50 events from security.log and then display the last 10 events from those 50.

```
(kali@kali)-[~/filters]
$ head -50 security.log | tail -10
2026-08-15 09:19:20 INFO user login successful; source=192.168.10.21; event_id=SEC0041
2026-08-15 09:19:49 WARN failed login attempt; source=192.168.10.22; event_id=SEC0042
2026-08-15 09:20:18 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0043
2026-08-15 09:20:47 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0044
2026-08-15 09:21:16 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0045
2026-08-15 09:21:45 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0046
2026-08-15 09:22:14 INFO user logout; source=203.0.113.77; event_id=SEC0047
2026-08-15 09:22:43 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0048
2026-08-15 09:23:12 INFO user login successful; source=192.168.10.21; event_id=SEC0049
2026-08-15 09:23:41 WARN failed login attempt; source=192.168.10.22; event_id=SEC0050
```


LINUX COMMANDS

bb. Display the last 50 events from security.log and then display the first 10 events from those 50.

```
(kali㉿kali)-[~/filters]
$ tail -50 security.log | head -10
2026-08-15 09:33:50 INFO user logout; source=203.0.113.77; event_id=SEC0071
2026-08-15 09:34:19 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0072
2026-08-15 09:34:48 INFO user login successful; source=192.168.10.21; event_id=SEC0073
2026-08-15 09:35:17 WARN failed login attempt; source=192.168.10.22; event_id=SEC0074
2026-08-15 09:35:46 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0075
2026-08-15 09:36:15 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0076
2026-08-15 09:36:44 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0077
2026-08-15 09:37:13 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0078
2026-08-15 09:37:42 INFO user logout; source=203.0.113.77; event_id=SEC0079
2026-08-15 09:38:11 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0080
```

cc. Display entries from line 11 to line 20 of login.log.

```
(kali㉿kali)-[~/filters]
$ sed -n '11,20p' login.log
Aug 15 09:10:10 server sshd[2111]: Failed password for root from 203.0.113.77
Aug 15 09:11:17 server sshd[2112]: Accepted password for guest from 192.168.10.45
Aug 15 09:12:24 server sshd[2113]: Accepted password for student1 from 192.168.10.21
Aug 15 09:13:31 server sshd[2114]: Failed password for student2 from 198.51.100.44
Aug 15 09:14:38 server sshd[2115]: Failed password for analyst from 203.0.113.77
Aug 15 09:15:45 server sshd[2116]: Accepted password for admin from 192.168.10.45
Aug 15 09:16:52 server sshd[2117]: Accepted password for root from 192.168.10.21
Aug 15 09:17:59 server sshd[2118]: Failed password for guest from 198.51.100.44
Aug 15 09:18:06 server sshd[2119]: Failed password for student1 from 203.0.113.77
Aug 15 09:19:13 server sshd[2120]: Accepted password for student2 from 192.168.10.45
```

dd. Display entries from line 21 to line 30 of login.log.

```
(kali㉿kali)-[~/filters]
$ sed -n '21,30p' login.log
Aug 15 09:20:20 server sshd[2121]: Accepted password for analyst from 192.168.10.21
Aug 15 09:21:27 server sshd[2122]: Failed password for admin from 198.51.100.44
Aug 15 09:22:34 server sshd[2123]: Failed password for root from 203.0.113.77
Aug 15 09:23:41 server sshd[2124]: Accepted password for guest from 192.168.10.45
Aug 15 09:24:48 server sshd[2125]: Accepted password for student1 from 192.168.10.21
Aug 15 09:25:55 server sshd[2126]: Failed password for student2 from 198.51.100.44
Aug 15 09:26:02 server sshd[2127]: Failed password for analyst from 203.0.113.77
Aug 15 09:27:09 server sshd[2128]: Accepted password for admin from 192.168.10.45
Aug 15 09:28:16 server sshd[2129]: Accepted password for root from 192.168.10.21
Aug 15 09:29:23 server sshd[2130]: Failed password for guest from 198.51.100.44
```

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ee. Display entries from line 11 to line 20 of access.log.

```
(kali㉿kali)-[~/filters]
$ sed -n '11,20p' access.log
192.168.10.30 - - [15/Aug/2026:09:06:10 +0530] "GET /admin HTTP/1.1" 404 1882
192.168.10.45 - - [15/Aug/2026:09:06:47 +0530] "GET /robots.txt HTTP/1.1" 401 2019
203.0.113.25 - - [15/Aug/2026:09:07:24 +0530] "GET /api/status HTTP/1.1" 200 2156
198.51.100.44 - - [15/Aug/2026:09:08:01 +0530] "POST /assets/app.js HTTP/1.1" 200 2293
203.0.113.77 - - [15/Aug/2026:09:08:38 +0530] "GET / HTTP/1.1" 302 2430
198.51.100.91 - - [15/Aug/2026:09:09:15 +0530] "GET /login HTTP/1.1" 403 2567
192.168.10.21 - - [15/Aug/2026:09:09:52 +0530] "GET /dashboard HTTP/1.1" 404 2704
192.168.10.22 - - [15/Aug/2026:09:10:29 +0530] "POST /admin HTTP/1.1" 401 2841
192.168.10.30 - - [15/Aug/2026:09:11:06 +0530] "GET /robots.txt HTTP/1.1" 200 2978
192.168.10.45 - - [15/Aug/2026:09:11:43 +0530] "GET /api/status HTTP/1.1" 200 3115
```

ff. Display entries from line 31 to line 40 of access.log.

```
(kali㉿kali)-[~/filters]
$ sed -n '31,40p' access.log
203.0.113.77 - - [15/Aug/2026:09:18:30 +0530] "GET /dashboard HTTP/1.1" 200 526
198.51.100.91 - - [15/Aug/2026:09:19:07 +0530] "GET /admin HTTP/1.1" 200 663
192.168.10.21 - - [15/Aug/2026:09:19:44 +0530] "GET /robots.txt HTTP/1.1" 302 800
192.168.10.22 - - [15/Aug/2026:09:20:21 +0530] "POST /api/status HTTP/1.1" 403 937
192.168.10.30 - - [15/Aug/2026:09:20:58 +0530] "GET /assets/app.js HTTP/1.1" 404 1074
192.168.10.45 - - [15/Aug/2026:09:21:35 +0530] "GET / HTTP/1.1" 401 1211
203.0.113.25 - - [15/Aug/2026:09:22:12 +0530] "GET /login HTTP/1.1" 200 1348
198.51.100.44 - - [15/Aug/2026:09:22:49 +0530] "POST /dashboard HTTP/1.1" 200 1485
203.0.113.77 - - [15/Aug/2026:09:23:26 +0530] "GET /admin HTTP/1.1" 302 1622
198.51.100.91 - - [15/Aug/2026:09:24:03 +0530] "GET /robots.txt HTTP/1.1" 403 1759
```

gg. Display security events from line 21 to line 30 of security.log.

```
(kali㉿kali)-[~/filters]
$ sed -n '21,30p' security.log
2026-08-15 09:09:40 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0021
2026-08-15 09:10:09 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0022
2026-08-15 09:10:38 INFO user logout; source=203.0.113.77; event_id=SEC0023
2026-08-15 09:11:07 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0024
2026-08-15 09:11:36 INFO user login successful; source=192.168.10.21; event_id=SEC0025
2026-08-15 09:12:05 WARN failed login attempt; source=192.168.10.22; event_id=SEC0026
2026-08-15 09:12:34 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0027
2026-08-15 09:13:03 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0028
2026-08-15 09:13:32 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0029
2026-08-15 09:14:01 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0030
```

hh. Display security events from line 51 to line 60 of security.log.

```
(kali@kali)-[~/filters]
$ sed -n '51,60p' security.log
2026-08-15 09:24:10 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0051
2026-08-15 09:24:39 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0052
2026-08-15 09:25:08 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0053
2026-08-15 09:25:37 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0054
2026-08-15 09:26:06 INFO user logout; source=203.0.113.77; event_id=SEC0055
2026-08-15 09:26:35 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0056
2026-08-15 09:27:04 INFO user login successful; source=192.168.10.21; event_id=SEC0057
2026-08-15 09:27:33 WARN failed login attempt; source=192.168.10.22; event_id=SEC0058
2026-08-15 09:28:02 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0059
2026-08-15 09:28:31 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0060
```

ii. Display incident events from line 101 to line 110 of incident.log.

```
(kali@kali)-[~/filters]
$ sed -n '101,110p' incident.log
2026-08-15 09:21:40 INFO login success; source=203.0.113.25; incident_event=101
2026-08-15 09:21:53 INFO web request; source=198.51.100.44; incident_event=102
2026-08-15 09:22:06 INFO logout; source=203.0.113.77; incident_event=103
2026-08-15 09:22:19 INFO scheduled scan; source=198.51.100.91; incident_event=104
2026-08-15 09:22:32 INFO access check; source=192.168.10.21; incident_event=105
2026-08-15 09:22:45 INFO login success; source=192.168.10.22; incident_event=106
2026-08-15 09:22:58 INFO web request; source=192.168.10.30; incident_event=107
2026-08-15 09:23:11 INFO logout; source=192.168.10.45; incident_event=108
2026-08-15 09:23:24 INFO scheduled scan; source=203.0.113.25; incident_event=109
2026-08-15 09:23:37 INFO access check; source=198.51.100.44; incident_event=110
```


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jj. Display incident events from line 201 to line 220 of incident.log.

```
(kali㉿kali)-[~/filters]
$ sed -n '201,220p' incident.log
2026-08-15 09:43:20 WARNING failed SSH login; source=192.168.10.21; incident_event=201
2026-08-15 09:43:33 ALERT firewall block; source=192.168.10.22; incident_event=202
2026-08-15 09:43:46 WARNING possible port scan; source=192.168.10.30; incident_event=203
2026-08-15 09:43:59 ALERT authentication failure; source=192.168.10.45; incident_event=204
2026-08-15 09:44:12 WARNING failed SSH login; source=203.0.113.25; incident_event=205
2026-08-15 09:44:25 ALERT firewall block; source=198.51.100.44; incident_event=206
2026-08-15 09:44:38 WARNING possible port scan; source=203.0.113.77; incident_event=207
2026-08-15 09:44:51 ALERT authentication failure; source=198.51.100.91; incident_event=208
2026-08-15 09:45:04 WARNING failed SSH login; source=192.168.10.21; incident_event=209
2026-08-15 09:45:17 ALERT firewall block; source=192.168.10.22; incident_event=210
2026-08-15 09:45:30 WARNING possible port scan; source=192.168.10.30; incident_event=211
2026-08-15 09:45:43 ALERT authentication failure; source=192.168.10.45; incident_event=212
2026-08-15 09:45:56 WARNING failed SSH login; source=203.0.113.25; incident_event=213
2026-08-15 09:46:09 ALERT firewall block; source=198.51.100.44; incident_event=214
2026-08-15 09:46:22 WARNING possible port scan; source=203.0.113.77; incident_event=215
2026-08-15 09:46:35 ALERT authentication failure; source=198.51.100.91; incident_event=216
2026-08-15 09:46:48 WARNING failed SSH login; source=192.168.10.21; incident_event=217
2026-08-15 09:47:01 ALERT firewall block; source=192.168.10.22; incident_event=218
2026-08-15 09:47:14 WARNING possible port scan; source=192.168.10.30; incident_event=219
2026-08-15 09:47:27 ALERT authentication failure; source=192.168.10.45; incident_event=220
```

kk. Display incident events from line 491 to line 500.

```
(kali㉿kali)-[~/filters]
$ sed -n '491,500p' incident.log
2026-08-15 10:46:10 INFO login success; source=192.168.10.30; incident_event=491
2026-08-15 10:46:23 INFO web request; source=192.168.10.45; incident_event=492
2026-08-15 10:46:36 INFO logout; source=203.0.113.25; incident_event=493
2026-08-15 10:46:49 INFO scheduled scan; source=198.51.100.44; incident_event=494
2026-08-15 10:47:02 INFO access check; source=203.0.113.77; incident_event=495
2026-08-15 10:47:15 INFO login success; source=198.51.100.91; incident_event=496
2026-08-15 10:47:28 INFO web request; source=192.168.10.21; incident_event=497
2026-08-15 10:47:41 INFO logout; source=192.168.10.22; incident_event=498
2026-08-15 10:47:54 INFO scheduled scan; source=192.168.10.30; incident_event=499
2026-08-15 10:48:07 INFO access check; source=192.168.10.45; incident_event=500
```

LINUX COMMANDS

ll. Take the first 20 entries of login.log and count how many lines are displayed.

```
(kali㉿kali)-[~/filters]
$ head -20 login.log | wc -l
20
```

mm. Take the last 15 entries of access.log and count how many lines are displayed.

```
(kali㉿kali)-[~/filters]
$ tail -15 access.log | wc -l
15
```

nn. Take the first 50 entries of security.log and count the number of words in those entries.

```
(kali㉿kali)-[~/filters]
$ head -50 security.log | wc -w
412
```

oo. Take the last 25 entries of security.log and count the number of lines.

```
(kali㉿kali)-[~/filters]
$ tail -25 security.log | wc -l
25
```

pp. Extract lines 21–30 from login.log and count the number of extracted lines.

```
(kali㉿kali)-[~/filters]
$ sed -e '21,30p' login.log | wc -l
60
```

qq. Extract lines 51–60 from access.log and count the number of extracted lines.

```
(kali㉿kali)-[~/filters]
$ sed -n '51,60p' access.log | wc -l
10
```

LINUX COMMANDS

rr. Extract events 101–150 from incident.log and count how many events are displayed.

```
(kali㉿kali)-[~/filters]
$ sed -n '101,150p' incident.log | wc -l
50
```

ss. Display the first 100 events from incident.log, take the last 20 events from that output, and count the number of lines.

```
(kali㉿kali)-[~/filters]
$ head -100 incident.log | tail -20 | wc -l
20
```